

McMaster University
SFWRENG 3DB3 Fall 2025
Assignment 2
Due: Oct 31, 2025 at 10:00pm

October 5, 2025

For this assignment, we will continue to work with *La Société de transport de Montréal (STM)*, and perform some data analytics. You will find the following files at this [link](#):

- a) `createTables_2025.ddl`: the DDL for CREATE TABLE statements
- b) `csv.zip`: containing the raw data files to import into your relations
- c) `import_csv.sh`: the IMPORT script
- d) `setup.pdf`: instructions on how to load the data into DB2.

Running this script can take a minimum of 10 - 20 mins, and possibly longer, depending on the server load. It is recommended that you do not wait until the last minute to populate your database!

These files correspond to a simplified schema shown in the E-R diagram `asg2ER.pdf`. Please run these scripts on your database (i.e., remember to update the CONNECT TO statement with your database name). You will use this schema for the questions below.

I. SQL (60 marks)

Write SQL statements for the following queries. Execute each of your SQL queries against your STM database, and give the result of each query.

- Q1) [6 marks] For each trip, compute the number of complaints. Return the trip ID only for trips with more than two complaints. Order the output in descending order by the number of complaints.
- Q2) [6 marks] Find all drivers who have at least one complaint about their conduct (complaint category = 'driver conduct') between November 1, 2024 to November 10, 2024. Return the driver's distinct PersonID, and name in ascending order of PersonID.

Q3) [6 marks] Count the number of distinct riders that travelled on each route between November 1, 2024 to December 31, 2024. Your output should include the route ID, route name, and the number of riders (as NumRiders). Display your output in descending order of NumRiders.

Q4) [7 marks] Find all bus routes (route name) that can be accessed from McGill subway station (i.e., McGill subway station's StopID = 'STATION_M140'). A bus stop is considered connected to McGill station if its SubwayStation attribute value equals 'STATION_M140'.

Note: Consider the schema:

Stops (StopID, Name, Type, WheelChairAccessibility, SubwayStation)

A stop describes either:

(1) a subway station (type = 1) and StopID will have a prefix 'STATION'; or

(2) a bus stop (type = 0) and a numeric StopID.

If a bus stop is located at a subway station, then its value of SubwayStation will equal the StopID of that subway station. Please see the FAQ for a specific example.

Q5) [7 marks] Find all riders who have not taken any trips in December 2024. Return their person ID (in ascending order by PersonID).

Q6) [8 marks] Find the names of all routes with at least one trip that does not run on both Saturday and Sunday. Return distinct route name.

Note: A Trip uses the ScheduleDays to determine which days a trip will operate. The (bitmap) schedule is an attribute where each of the low 7-bits (64, 32, 16, 8, 4, 2, 1) represents whether service runs on Mon, Tue, Wed, Thu, Fri, Sat, and Sun respectively (1: service, 0: no service).

- Example 1: A schedule 96 (= 0b1100000 = 64 + 32) indicates the trip only operates on Monday and Tuesday.
- Example 2: If a trip operates with service on Saturday and Sunday, it follows a Schedule-Days with schedule = 0b0000011 = 2 + 1 = 3).
- Hint: The operator '&' represents the bitwise 'AND' operation (returns 1 only if both operands are 1), and the operator '|' represents the bitwise 'OR' operation (returns 1 if at least one operand is 1).
 - Example 3: 96 (=0b1100000) & 127 (=0b1111111) = 96 (=0b1100000)
 - Example 4: 96 (=0b1100000) | 127 (=0b1111111) = 127 (=0b1111111)

Q7) [10 marks] A complaint with category 'overcrowding' was filed by rider 1313 on '2024-12-18'. To investigate this matter, return the person's name filing the complaint, timestamp, category of the complaint, driver's name and license number, route name, and trip direction. Your output schema should contain the following re-named attributes, respectively, RiderName, ComplaintTime, ComplaintCategory, DriverName, LicenseNo, RouteName, Direction.

Q8) [10 marks] For a driver performance evaluation from '2024-10-28' to '2024-12-31', write a query to find each driver's total number of trips, and the total number of passengers they each transported. Only trips on weekdays (Monday–Friday) that carried at least one passenger should

be included. Please also only include drivers with a total passenger count of at least 800 (inclusive). The `ScheduleDays` table defines when trips run, and it is assumed that all scheduled trips were completed. Return the driver name, the total number of trips, and the total number of passengers, renamed to `DriverName`, `TotalTrips`, `TotalPassengers`, respectively. Display your results in descending order of `TotalTrips`.

Hint 1: Weekday schedules are identified by `schedulebitmap = 124`.

Note: You must find all schedules that were active at any point during the evaluation period. This means you are looking for any schedule whose validity period **overlaps** with the evaluation period, not just those fully contained within it. For example, a schedule running from '2024-10-01' to '2024-11-30' must be included.

II. Relational Algebra (48 marks)

For the queries Q1 - Q8 in Part I, give the corresponding relational algebra expression. Please submit your **typed** answers, using your choice of software (e.g., LaTeX, MS Word). **Handwritten relational algebra solutions will not be accepted.**

III. Indexes (10 marks)

For the workload given in Part I, several queries have been showing poor performance (i.e., increasing response times). Your task is to improve the performance of this workload as much as possible by recommending two (non-primary key) indexes that should be defined on any of the tables. What two indexes would you recommend? Indexes that contain only the primary key will not be accepted. For each index, state:

- The attribute(s) the index is defined on.
- Properties of the index (e.g., type of index, clustered/unclustered, composite, etc.)
- Which queries (Q1 - Q8) you think this index will help, and why.

Grading

This assignment is worth 14% towards your final grade.

Submission

All files are to be submitted using Crowdmark. Please ensure you submit all files with the correct names, as described below. In each file, include your name and student ID number. Upload four files with the indicated file extensions (no compression based `.tar`, `.zip`, `.rar` files will be accepted).

- For Part I:

- Submit your SQL statements and the result for each query in two files. Submit your SQL statements in a script file called `queries.sql`, and the corresponding query results in a file called `queries.results`.
 - Ensure your SQL statements are syntactically correct and that they are executable on the DB2 server. **Non-executable queries (that do not run on the DB2 server) will not be marked, and will receive a grade of zero. If there is no “connect to SE3DB3” statement, marks will be deducted.**
 - Clearly label, with comments, which query the result tuples correspond to in `queries.results`.
- For Part II: Submit your relational algebra expressions in a file named `ra.pdf`.
 - For Part III: Submit your index recommendations in a file named `index.pdf`.